



ElectronicsTutorials

DC CIRCUIT THEOREMS

For Students, Professionals
and Beyond

eBook 3

ASPENCORE

WWW.ELECTRONICS-TUTORIALS.WS

TABLE OF
CONTENTS

1. Introduction	1
2. Ohms Law	1
3. Resistors in Series	3
4. Series Resistor Voltage Drops	3
5. Resistors In Parallel	4
6. Kirchhoff's Circuit Laws.	5
7. Mesh Current Analysis	7
8. Nodal Voltage Analysis	8
9. Thevenin's Theorem	9
10. Norton's Theorem	11
11. Maximum Power Transfer Theorem	13
13. Star/Delta and Delta/Star Transformations	14

Our Terms of Use

This **Basic Electronics Tutorials eBook** is focused on the DC circuit theorems with the information presented within this ebook provided “as-is” for general information purposes only.

All the information and material published and presented herein including the text, graphics and images is the copyright or similar such rights of Aspencore. This represents in part or in whole the supporting website: **www.electronics-tutorials.ws**, unless otherwise expressly stated.

This free e-book is presented as general information and study reference guide for the education of its readers who wish to learn Electronics. While every effort and reasonable care has been taken with respect to the accuracy of the information given herein, the author makes no representations or warranties of any kind, expressed or implied, about the completeness, accuracy, omission of errors, reliability, or suitability with respect to the information or related graphics contained within this e-book for any purpose.

As such it is provided for personal use only and is not intended to address your particular problem or requirement. Any reliance you place on such information is therefore strictly at your own risk. We can not and do not offer any specific technical advice, troubleshooting assistance or solutions to your individual needs.

We hope you find this free publication useful and enlightening. For more information about any of the topics covered herein please visit our online website at:

www.electronics-tutorials.ws

1. INTRODUCTION

Individual circuit elements can be interconnected together in a number of different ways to provide either single or multiple paths for current to flow around a circuit. Commonly, each circuit element is defined by its own electrical characteristics depending upon what the circuit element is. For example, a resistor is defined by its resistance, a capacitor is defined by its capacitance, and an inductor is defined by its inductance.

For a simple linear circuit, we can use basic mathematics to find the voltages and currents flowing around each branch. But for more complex circuits with multiple circuit elements and/or parallel branches we require something a little more involved. Thus allowing us to find an equivalent circuit or component which will reproduce exactly the same electrical behaviour of the circuit or component it is intended to replace.

For example, we can replace two resistors connected in series with one single equivalent resistor giving: $R_{EQ} = R_1 + R_2$, or the voltages of two independent voltage sources V_1 and V_2 with one single equivalent voltage source representing V .

Whether the circuit to be analysed is simple, complex, connected in a series, parallel or configurations of both, or whether it just consists of a single battery voltage source supplying a resistance, we can analyse the vast majority of them all using the generalised equations of Ohm's Law.

For example, if we apply a constant voltage (V), at some point in a circuit and measure the current (I), flowing past that point, we can determine its resistance (R), simply using the equation that: $R = V/I$.

However, for more complex interconnected circuits that use bridge or T networks, or when capacitors and inductors are included, then we cannot simply use Ohm's Law alone to find the voltages, currents or component values within the circuit.

For these types of calculations, we need to apply and follow certain rules, law's and transformations which would allow us to analyse the circuit more easily to find the equivalent values of interest to us.

2. OHMS LAW

Ohm's Law is the most basic, simplest and easiest law to use for circuit analysis. It describes the definitive relationship between voltage, current, and resistance in an electrical circuit. Whether it is supplied from a battery source, with a constant DC voltage and/or current, or from a sinusoidal AC source using RMS (root mean squared) values.

German physicist Georg Simon Ohm, conducted many laboratory experiments during his life time. But in one of them he found that, at a constant temperature, the electrical current flowing through a fixed linear resistance is directly proportional to the voltage developed across it.

Ohm's Law states that the current flowing through a resistor is directly proportional to the voltage across it

But he also observed that the same circuit current is inversely proportional to the value of the resistance. That is, the circuit current varies if the resistance varies while the voltage remains constant.

This relationship between the Current, Voltage and Resistance therefore forms the bases of Ohm's Law and is shown below in Equation 1.

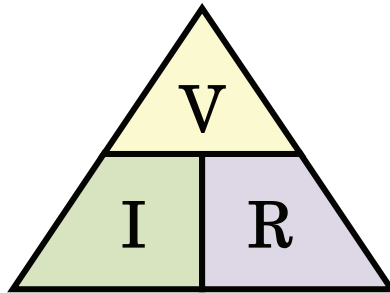
EQUATION 1. OHM'S LAW RELATIONSHIP

$$\text{Current (I)} = \frac{\text{Voltage (V)}}{\text{Resistance (R)}} \text{ in Amperes (A)}$$

Thus by knowing any two of the three possible values of the Voltage, Current or Resistance quantities we can use Ohm's Law to find the third missing value. Ohm's Law is used extensively in electronics formulas and calculations so it is "very important to understand and accurately remember these formulas".

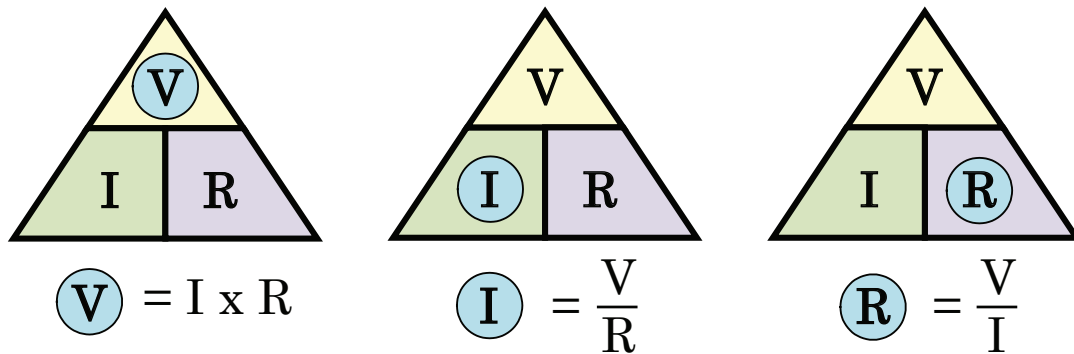
We can superimpose the three electrical quantities of V, I and R into a triangular format (affectionately called the Ohm's Law Triangle) giving voltage at the top with current and resistance below. This arrangement represents the actual position of each quantity within the respective Ohm's law formulas, as shown in Figure 1.

FIGURE 1. OHM'S LAW TRIANGLE



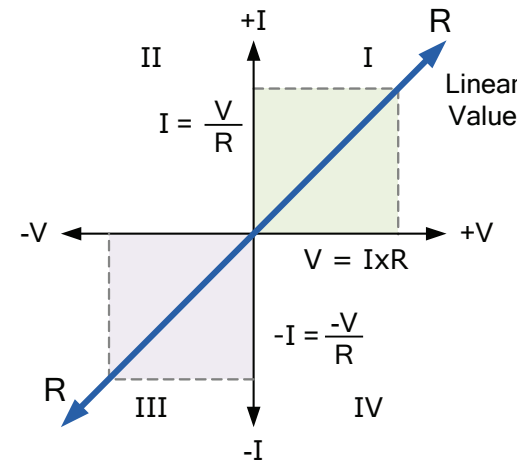
Transposing the previous Ohm's Law equation above, we can produce the following individual triangles of the same standard equation as shown in Figure 2.

FIGURE 2. INDIVIDUAL TRIANGLE EQUATIONS



A resistance (or resistor) which obeys the above rules of Ohm's law is known as a linear resistance. Linear resistors have a constant resistance for all values of positive or negative voltages and currents.

FIGURE 3. LINEAR RESISTANCE I-V CHARACTERISTICS



The relationship between voltage, (V) and current, (I) in a linear circuit of constant resistance, (R) would produce a straight line I-V characteristics curve with the angle of its slope equal to the value of the resistance as shown in Figure 3.

Electrical resistance R, is the coefficient of proportionality between the voltage and current, as current (I) is simply proportional to voltage (V) (positive or negative).

This therefore allows any quantity to be easily found whenever two of the three quantities are known.

Resistors are classed as being passive circuit elements, and as such they cannot deliver power to a circuit or store energy. Unlike a capacitor which stores energy in the form of an electrostatic field, or an inductor that stores its energy in the form of a magnetic field.

Instead resistors, of either fixed or variable value absorb power that appears externally as either heat and light. While resistors can be used in AC and DC circuits to limit current or providing voltage distribution. The power dissipated in a resistance will always be positive regardless of voltage polarity and current direction.

This linear relationship gives a current-voltage (I-V) characteristic of a straight line which also includes metal conductors. Thus they are classed as Ohmic elements obeying Ohm's law as demonstrated in Figure 3.

Circuit elements such as diodes, transistors, photo-devices, lamp filaments, etc. are nonlinear devices which do not obey Ohm's law. They generally have a voltage drop across them that is proportional to some power of the current and therefore have a non-straight I-V characteristics curve. Thus they are classed as Non-Ohmic elements.

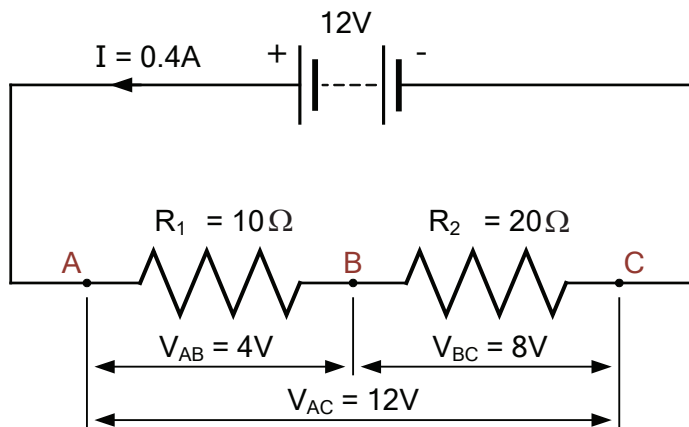
3. RESISTORS IN SERIES

The most basic and fundamental circuit analysis we can perform, is that of two resistors in series. Resistors are said to be connected together in “Series”, when they are daisy chained together to form one single line through which electric current can flow.

Since all the current flowing through the first resistor has no other way to go it must therefore pass through the second resistor, and the third and so on within the series chain as shown in Figure 4.

Thus resistors in series have a Common Current flowing through them because the same current that flows through one resistor must also flow through the others as it can only take one path. Then: $I_{R1} = I_{R2} = I_{R3} = I_{R4}$ etc.

FIGURE 4. SERIES RESISTORS



As the resistors are connected together in series the total resistance, R_T of the circuit must be equal to the mathematical sum of all the individual resistors added together. That is:

$$R_T = R_1 + R_2 = 10\Omega + 20\Omega = 30\Omega$$

Total resistance is defined as: “a single value of resistance that can replace any number of resistors in series without altering the values of the current or voltages in the circuit”. Thus we could replace the two individual resistors above with just one single “equivalent” resistor R_{EQ} which would have a resistive value of 30Ω between points A and C.

When three, four, five or even more resistors are all connected together in a series circuit, the total or equivalent resistance of the circuit, R_{EQ} would still be the mathematical sum of all the individual resistors connected together.

The more resistors added to a series chain, the greater the total resistance no matter their value.

Note that the equivalent resistance (R_{EQ}) of any two or more resistors connected together in series will always be GREATER than the value of the largest resistor in the series chain. In our example above, $R_{EQ} = 30\Omega$ whereas the largest value resistor is only 20Ω .

Series resistors always have a common current. The equivalent resistance will always be greater than the largest resistive value

If the two resistances connected together in series are of equal value, then the total or equivalent resistance, R_{EQ} will be equal to twice the value of one of the resistors. That is, R_{EQ} will be equal to $2R$ and for three equal resistors in series, $3R$, etc. Otherwise the equivalent resistance is calculated as: $R_1 + R_2 + R_3 + R_4 + \dots$ etc.

4. SERIES RESISTOR VOLTAGE DROPS

The voltage dropped across series connected resistors follows different rules to that of the series current. The supply voltage (V) is divided proportionally across the resistors relative to their resistive value and the series current flowing through them. The sum of all the voltage drops across each series resistor will be equal to the supply voltage.

Thus for our previous series resistor example in Figure 4.

$$V_{AC} = V_{AB} + V_{BC} = 4V + 8V = 12V$$

Using the second Ohm's Law triangle, the current flowing around the series circuit is calculated as:

$$I = \frac{V}{R_T} = \frac{12}{30} = 0.4\text{A or } 400\text{mA}$$

Now using the first Ohm's Law triangle, the voltage across the individual resistors of Figure 4 can be calculated as:

$$\text{Voltage across } R_1 = V_{AB} = I \times R_1 = 0.4\text{A} \times 10\Omega = 4\text{V}$$

$$\text{Voltage across } R_2 = V_{BC} = I \times R_2 = 0.4\text{A} \times 20\Omega = 8\text{V}$$

Thus giving a total voltage V_{AC} of: $4\text{V} + 8\text{V} = 12\text{V}$ which is equal to the value of the supply voltage. Then the sum of the potential differences across the resistors is equal to the total potential difference across the combination and in our example this is 12V.

So we can see that series resistors divide the applied voltage proportionally between themselves. Therefore series resistor networks can also be thought of as "voltage dividers" since they split the supply voltage proportionally across each resistor in the series chain.

Thus a series resistor circuit having "N" resistive components will have N-different voltages across it while maintaining a common current. By using Ohm's Law, either the voltage, current or resistance of any series connected circuit can easily be found.

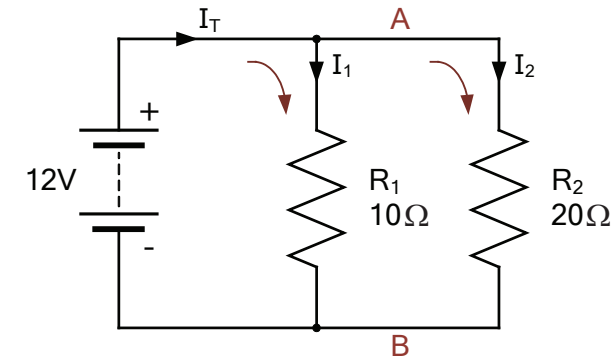
5. RESISTORS IN PARALLEL

Resistors are said to be connected together in "parallel", when both of their connecting terminals are respectively connected to the terminals of the other resistor or resistors.

Also in a parallel network the circuit current can take more than one path as there are multiple paths or branches for the current to flow through.

Therefore, the current is not the same at all points in a parallel circuit, and as such parallel circuits are classified as "current dividers".

FIGURE 5. PARALLEL RESISTORS



The voltage across all of the components in a parallel network IS always the same. That is resistors in parallel have a "common voltage" across them and this is true for all parallel connected elements.

In the parallel resistor example of Figure 5. The voltage across resistor R_1 equals the voltage across resistor R_2 which equals the supply voltage. Thus: $V_{AB} = V_{R1} = V_{R2} = 12\text{V}$.

Parallel Resistors always have a common voltage across them. The equivalent resistance will always be smaller than the lowest resistive value

For the previous series resistive network, we saw that the total resistance, R_T of the circuit was equal to the sum of all the individual resistors added together. For resistors in parallel the equivalent circuit resistance R_{EQ} is calculated differently.

Here, the reciprocal ($1/R$) value of the individual resistances are all added together instead of the resistances themselves. The inverse of the algebraic sum of all the reciprocal values added together from $1/R_1$ to $1/R_N$ gives the total resistance, R_T of the parallel circuit as shown.

$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_N}}$$

So the inverse of the equivalent resistance of two or more resistors connected in parallel is the algebraic sum of the inverses of the individual resistances. For our simple parallel circuit above in Figure 5, the total resistance is calculated as:

$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{1}{\frac{1}{10} + \frac{1}{20}} = \frac{1}{0.15} = 6.67 \Omega$$

Using the second Ohm's Law triangle, the total current flowing around the parallel circuit is calculated as:

$$I = \frac{V}{R_T} = \frac{12}{6.67} = 1.8 \text{ A}$$

Again using the Ohm's Law triangle, the currents flowing through the two resistors are calculated as:

$$\text{The current through } R_1 = \frac{V}{R_1} = \frac{12}{10} = 1.2 \text{ Ampere}$$

$$\text{The current through } R_2 = \frac{V}{R_2} = \frac{12}{20} = 0.6 \text{ Ampere}$$

Thus giving a total supply current I_T of: $1.2\text{A} + 0.6\text{A} = 1.8\text{A}$ as calculated previously.

This reciprocal method of calculation can be used for calculating any number of individual resistances connected together in parallel. If however, there are only two individual resistors in parallel then a much simpler and quicker formula can be used to find the total resistance value, and this is given as:

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{10 \times 20}{10 + 20} = \frac{200}{30} = 6.67 \Omega$$

The same 6.67Ω as for the longer reciprocal method. Also, if the two resistances (or impedances) in parallel are equal and of the same ohmic value, ($R_1 = R_2$) then the total or equivalent resistance, R_T is equal to half the value of one resistor. That is equal to $R/2$ and for three equal value resistors in parallel, $R/3$, etc.

Note also that the equivalent resistance will always be less than the smallest resistor in the parallel network. Then the total resistance, R_T will always decrease as additional parallel resistors are added.

6. KIRCHHOFF'S CIRCUIT LAWS

Previously we have seen that a single equivalent resistance, (R_T) can be found when two or more resistors are connected together in either series, parallel or combinations of both, and that these circuits obey Ohm's Law.

We also saw that components connected together in Series have the same current value flowing through them, and that components connected together in Parallel have the same voltage applied across them.

Kirchhoff's current and voltage laws can be used to analyse complex circuits

However, sometimes in complex electrical circuits and networks, we cannot simply use Ohm's Law alone to find the voltages or currents circulating within the circuit. For these types of calculation's, we need certain rules which allow us to obtain the circuit equations and for this we can use Kirchhoff's Circuit Laws.

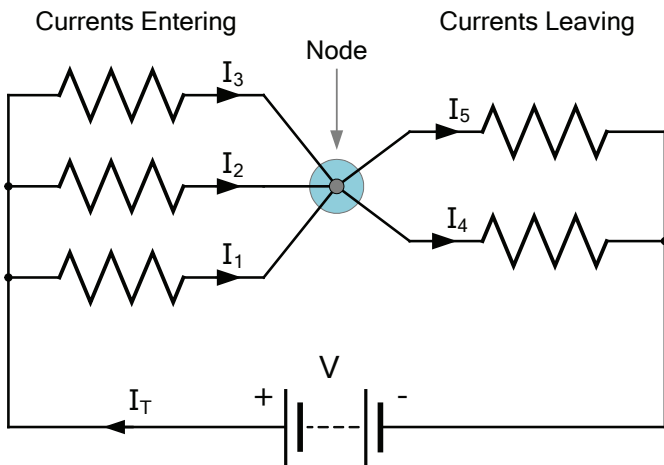
In 1845, a German physicist, Gustav Kirchhoff developed a set or pair of rules, laws or equations which deal with the conservation of current and energy around an electrical circuit. One of Kirchhoff's laws deals with the current flowing around a closed circuit, called Kirchhoff's Current Law, (KCL). The other law deals with the voltage sources present in a closed circuit and is called Kirchhoff's Voltage Law, (KVL).

6.1 Kirchhoff's First Law – The Current Law (KCL)

Kirchhoff's Current Law or KCL, deals with the conservation of electric charge and states that the “total current entering a junction or node is exactly equal to the currents leaving the node as they have no other place to go and no current is lost within the node”.

In other words, the algebraic sum of ALL the currents entering a node and leaving the same node (Note: a node is a circuit junction where two or more elements are electrically joined) must be equal to zero as: $I_{(ENTERING)} - I_{(LEAVING)} = 0$.

FIGURE 6. KIRCHHOFF'S CURRENT LAW



Here, the three currents entering the node, I_1 , I_2 , I_3 are all positive in value as they are entering. The two currents leaving the node, I_4 and I_5 are negative in value as they are leaving. Then this means we can also rewrite the same equation as:

$$\sum I_{\text{ENTERING}} = \sum I_{\text{LEAVING}}$$

$$I_1 + I_2 + I_3 = I_4 + I_5$$

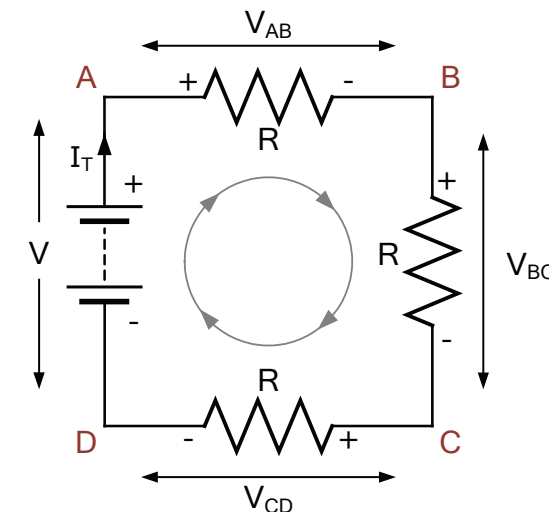
$$\therefore I_1 + I_2 + I_3 - I_4 - I_5 = 0$$

That is the sum of the currents entering and leaving the same node will be equal to zero.

6.2 Kirchhoff's Second Law – The Voltage Law (KVL)

Kirchhoff's Voltage Law or KVL, deals with the conservation of electrical energy and states that “in any closed loop or network, the total voltage around the loop is equal to the sum of all the voltage drops and rises within the same loop” as shown in Figure 7.

FIGURE 7. KIRCHHOFF'S VOLTAGE LAW



In other words, if we go around a closed loop in a specified and constant direction, either clockwise or anticlockwise, the algebraic sum of all the voltage drops around the loop exactly equals the sum of all the voltage rises within the same loop being equal to zero.

The voltage source (V) is driving a current I_T around the circuit. The sum of the $I_T \times R$ voltage drops within the loop equals the voltage source. That is:

$$\sum(ABCD) = V = V_{AB} + V_{BC} + V_{CD}$$

$$V - (V_{AB} + V_{BC} + V_{CD}) = 0$$

$$\therefore +V - V_{AB} - V_{BC} - V_{CD} = 0$$

Not that you can start at any point or node within a loop, but you must continue in the same direction noting the polarity of all the voltage drops, either positive or negative, and return back to the same starting point or node. It's important to maintain the same direction, clockwise or anti-clockwise or the final voltage sum will not be equal to zero.

7. MESH CURRENT ANALYSIS

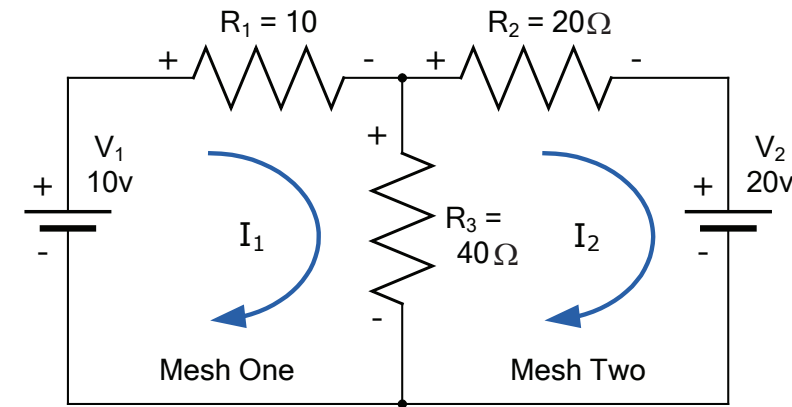
Any linear circuit which contains resistances and independent voltage sources can be solved using Mesh Current Analysis or simply "loop analysis". For circuit analysis, a mesh is a single closed loop path which does not contain any other paths or loops inside it, allowing us to label each "closed loop" with a circulating current. (I_1, I_2, I_3 , etc.)

Using Kirchhoff's voltage law (KVL) we can solve for all unknown mesh currents labelling all loops in a clockwise direction covering all circuit elements at least once. First we must identify mesh (loops) and assign a current to each mesh.

After which we can apply Kirchhoff's voltage law around each loop to get an equation in terms of the two loop currents, the current through load resistor R_3 and the voltage drop across R_3

The circuit of Figure 8. has two voltage sources. For each voltage source, current is assumed to flow in the direction as shown.

FIGURE 8. MESH CURRENT ANALYSIS



Apply Kirchhoff's Voltage Law to Mesh One

$$[V_1] = [I_1] \times [R]$$

$$10 - [I_1 10\Omega + (I_1 - I_2) 40\Omega] = 0$$

$$10 = I_1 10\Omega + (I_1 - I_2) 40\Omega$$

Apply Kirchhoff's Voltage Law to Mesh Two

$$[V_2] = [I_2] \times [R]$$

$$20 - [I_2 20\Omega + (I_2 - I_1) 40\Omega] = 0$$

$$-20 = I_2 20\Omega + (I_2 - I_1) 40\Omega$$

Combining the two equations into a single matrix/vector equation gives:

$$\begin{bmatrix} 10 \\ -20 \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \times \begin{bmatrix} 50 & -40 \\ -40 & 60 \end{bmatrix}$$

The Inverse of $\mathbf{R} = \begin{bmatrix} 50 & 40 \\ 40 & 60 \end{bmatrix}$ therefore $R^{-1} = (60 \times 50) - (40 \times 40) = 1400$

$$I_1 = \frac{(60 \times 10) + (40 \times -20)}{1400} = -0.143 \text{ A}$$

$$I_2 = \frac{(40 \times 10) + (50 \times -20)}{1400} = -0.429 \text{ A}$$

$$I_3 = I_1 - I_2 = -0.143 - (-0.429) = \underline{0.286 \text{ A}}$$

Note that the negative sign given for I_1 indicates that the clockwise direction of current flow initially chosen was wrong, but never the less still valid. In fact, the 20v battery is charging the 10v battery which we would expect as it is the greater voltage source.

Finally, the voltage drop across R_3 is calculated as:

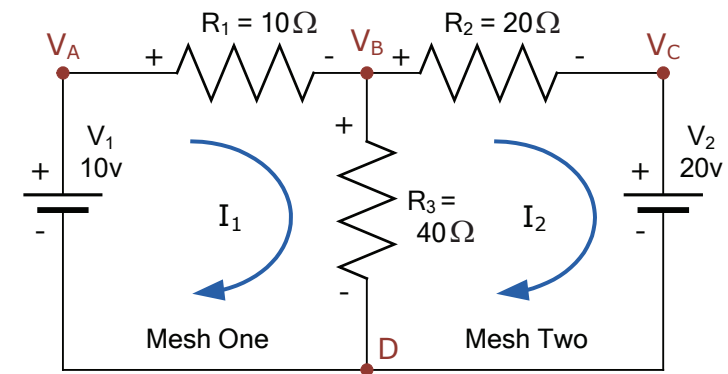
$$V_3 = I_3 \times R = 0.286 \times 40 = \underline{11.44 \text{ Volts}}$$

Thus the voltage developed across resistor R_3 is 11.44 volts.

8. NODAL VOLTAGE ANALYSIS

Nodal Voltage Analysis is the opposite to the previous mesh analysis and is based on the same concepts of matrix analysis. As its name implies, Nodal Voltage Analysis uses the “Nodal” equations of Kirchhoff’s current law (KCL) to find the voltage potentials around a circuit. Consider the circuit of Figure 9 from the previous mesh current analysis example.

FIGURE 9. NODAL VOLTAGE ANALYSIS



First we must select one node as the reference node. Node D is chosen as the reference node and the other three are referenced with respect to the reference node. They are assumed to have voltages: V_A , V_B and V_C with respect to node “D”.

Next we apply Kirchhoff’s current law to each non reference node and use Ohm’s law to express the branch currents in terms of node voltages. Solving as before to obtain the unknown node voltage at V_B and the current through load resistor R_3

As the voltage at: $V_A = 10\text{V}$ and the voltage at: $V_C = 20\text{V}$. Therefore:

$$I_{R3} = I_{R1} + I_{R2}$$

$$\therefore \frac{V_B}{40} = \frac{(V_A - V_B)}{10} + \frac{(V_C - V_B)}{20}$$

Convert resistance (R) to admittance (R^{-1}) matrix. Then the voltage at node V_B equals:

$$\frac{V_B}{40} = \frac{(10 - V_B)}{10} + \frac{(20 - V_B)}{20} = \left(\frac{10}{10} - \frac{V_B}{10} \right) + \left(\frac{20}{20} - \frac{V_B}{20} \right)$$

$$\frac{V_B}{40} = \left(1 - \frac{V_B}{10} \right) + \left(1 - \frac{V_B}{20} \right)$$

$$\text{Thus: } 2 = V_B \left(\frac{1}{40} + \frac{1}{10} + \frac{1}{20} \right) = V_B \times 0.175$$

$$V_B = 2 \left(\frac{1}{0.175} \right) = \frac{2}{0.175} = \underline{\underline{11.44 \text{ Volts}}}$$

$$\text{Therefore: } I_3 = \frac{V_B}{R_3} = \frac{11.44}{40} = 0.286 \text{ Amperes}$$

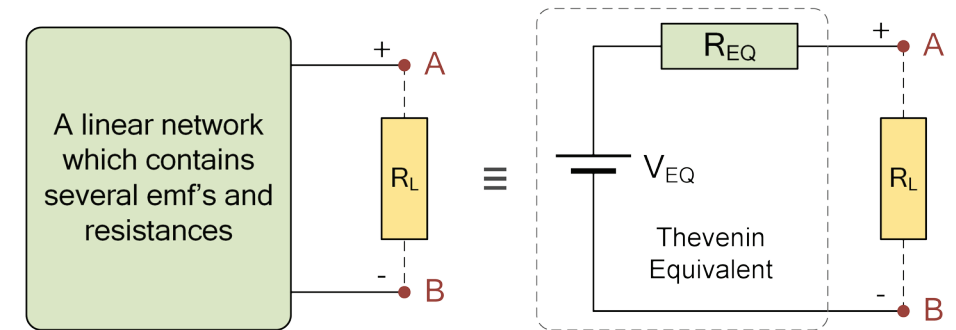
The same current value as calculated for the mesh current analysis.

9. THEVENIN'S THEOREM

Thevenin's theorem is another analytical method we can use to change a complex circuit into a simple equivalent circuit consisting of a single resistance in series with a source voltage. Thevenin's Theorem states that **"Any linear circuit containing several voltages and resistances can be replaced by just one single voltage in series with a single resistance connected across the load"**.

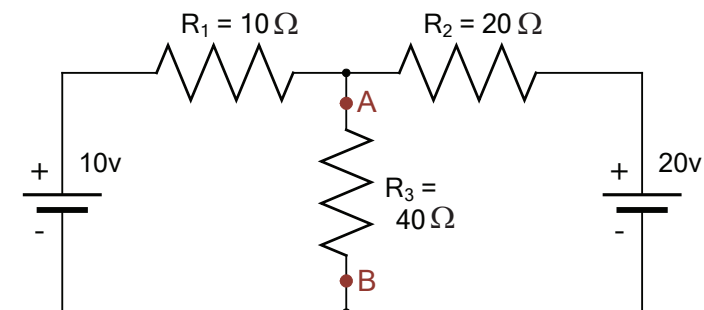
In other words, it is possible to simplify any electrical circuit, no matter how complex, to an equivalent two-terminal circuit consisting of just one equivalent independent or dependent voltage source (V_{EQ}) in series with a single equivalent resistance (R_{EQ}) connected to a load as shown in Figure 10.

FIGURE 10. THEVENIN EQUIVALENT CONFIGURATION



Where: R_{EQ} is the equivalent resistance value looking back into the circuit and V_{EQ} is the open circuit voltage across the terminals A and B. Consider the circuit of Figure 11 from the previous loop analysis examples.

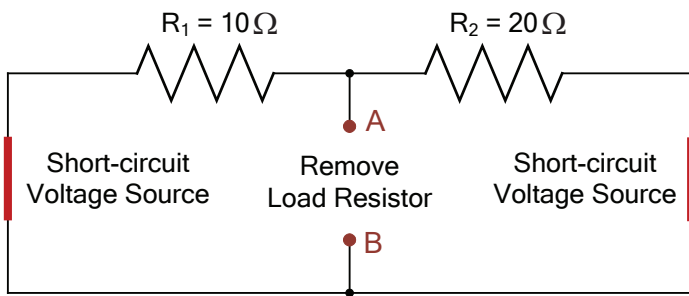
FIGURE 11. THEVENIN CIRCUIT



To find the equivalent Thevenin circuit we must first make two assumptions. No1. All voltage sources are replaced with a short circuit condition, thus making them equal to zero voltage, 0V (zero resistance). No2. All current sources are replaced with an open circuit condition, thus making them equal to zero current, 0A (infinite resistance).

To determine the equivalent resistance, R_{EQ} , we must first remove the 40Ω load resistance and short circuit the voltage sources as shown in Figure 12.

FIGURE 12. EQUIVALENT RESISTANCE, R_{EQ}

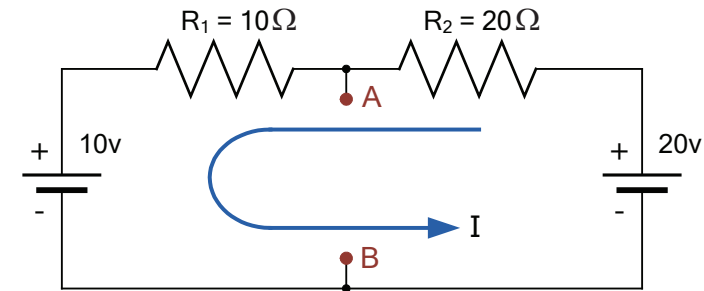


Resistor R_1 is now effectively in parallel with resistor R_2 so we can use the simpler equation above as there are only two resistors in parallel.

$$R_{EQ} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{10 \times 20}{10 + 20} = \frac{200}{30} = 6.67 \Omega$$

Then the equivalent parallel resistance, R_{EQ} is calculated as being: 6.67Ω. To determine the output voltage V_{EQ} across the terminals A and B when there is an open circuit condition, we now need to reconnect the two voltage sources back into the circuit, with the 40Ω load resistance still removed as shown in Figure 13.

FIGURE 13. EQUIVALENT OPEN CIRCUIT VOLTAGE, V_{EQ}



We can now use Ohm's Law to find the value of the circulating current, I in Figure 13.

$$I = \frac{V}{R} = \frac{20 - 10}{20 + 10} = \frac{10}{30} = 0.333 \text{ Amperes}$$

As the circulating current of 0.33 amperes (330mA) is common to both resistors, R_1 and R_2 , the voltage drop across each resistor can be calculated as:

$$V_{AB} \text{ (anticlockwise)} = V_2 - IR_2 = 20 - (20 \times 0.333) = 13.33 \text{ volts}$$

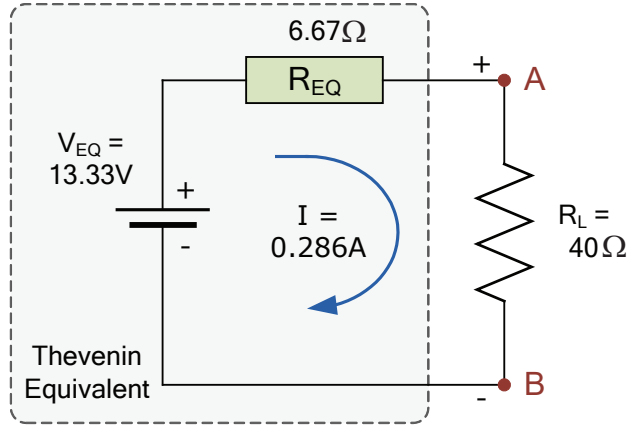
and / or

$$V_{AB} \text{ (clockwise)} = V_1 + IR_1 = 10 + (10 \times 0.333) = 13.33 \text{ volts}$$

Thus the Thevenin's circuit would consist of an equivalent resistance, $R_{EQ} = 6.67\Omega$ and an open circuit equivalent voltage source, $V_{EQ} = 13.33$ volts.

We can now connect the 40Ω resistor back into the circuit as shown in Figure 14.

FIGURE 14. THEVENIN EQUIVALENT CIRCUIT



Therefore:
$$I_L = \frac{V_{EQ}}{R_{EQ} + R_L} = \frac{13.33}{6.67 + 40} = 0.286 \text{ Amperes}$$

Again, the same current value as for the previous circuit analysis examples.

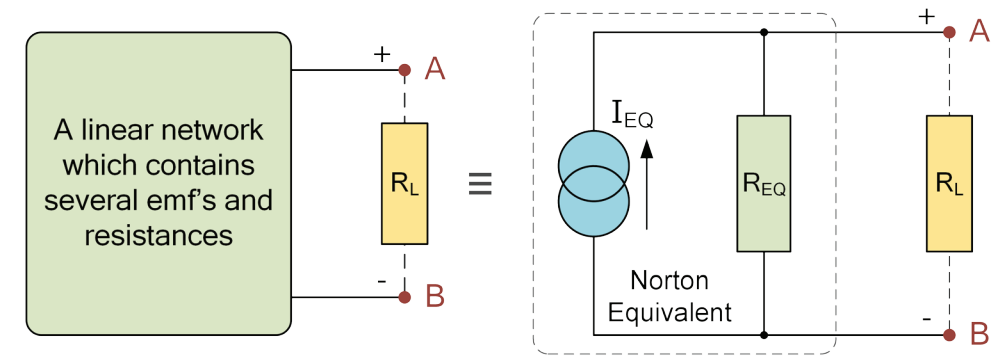
10. NORTON'S THEOREM

Thevenin's theorem reduces a complex network into an equivalent circuit called a voltage source. Where: V_{EQ} is the open-circuit voltage source, and R_{EQ} is the internal source resistance looking across the terminals A and B.

Norton's theorem reduces a network into an equivalent constant current source. Where I_{EQ} is the current source and R_{EQ} is the internal source resistance looking across the terminals A and B.

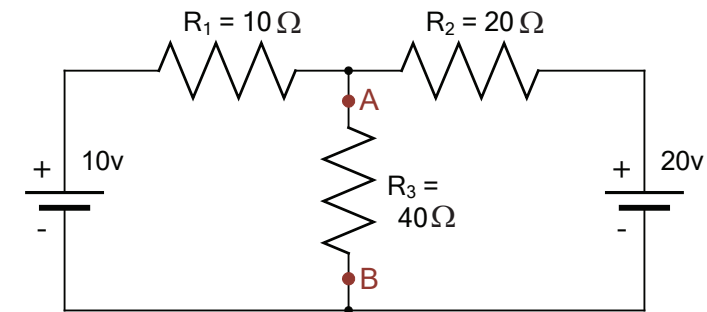
As Norton's Theorem states that **“Any linear circuit containing several energy sources and resistances can be replaced by a single Constant Current generator in parallel with a Single Resistor”** and this idea is shown in Figure 15.

Figure 15. Norton Equivalent Configuration



Where: R_{EQ} is the equivalent resistance value looking back into the circuit and I_{EQ} is the current source across the terminals A and B. Consider the circuit of Figure 16 from the previous loop analysis examples.

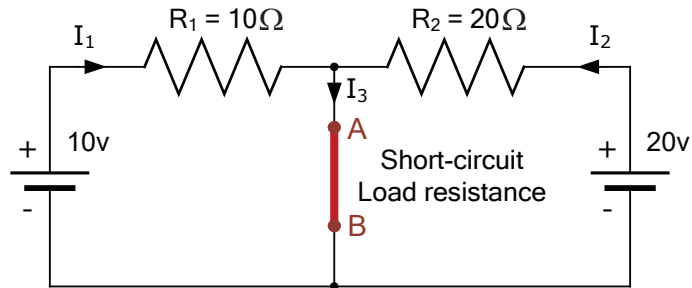
FIGURE 16. NORTON CIRCUIT



As with the previous Thevenin circuit, we must first make the same two assumptions. No1. All the voltage sources are replaced with a short circuit condition. No2. All the current sources are replaced with an open circuit condition.

To determine the equivalent current source, I_{EQ} , for Norton analysis, we must first remove the 40Ω load resistance and short circuit the terminals A and B as shown in Figure 17.

FIGURE 17. EQUIVALENT CURRENT SOURCE, I_{EQ}



With terminals A-B shorted, the two resistors are connected across their two respective voltage sources and the currents flowing through each resistor as well as the total short circuit current, I_3 is calculated as:

$$I_1 = \frac{V_1}{R_1} = \frac{10}{10} = 1 \text{ Ampere} \quad \text{and} \quad I_2 = \frac{V_2}{R_2} = \frac{20}{20} = 1 \text{ Ampere}$$

Therefore:

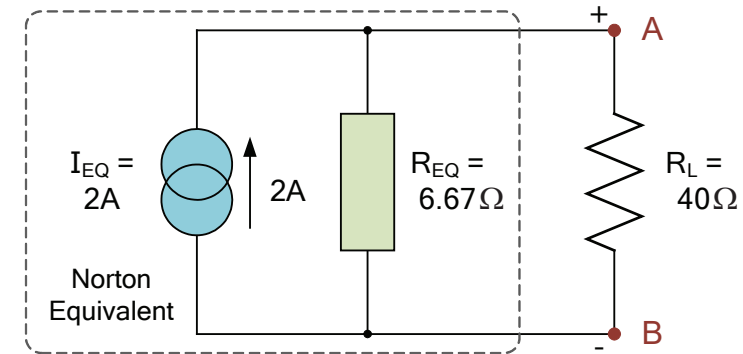
$$I_3 = I_1 + I_2 = 1 + 1 = 2 \text{ Amperes}$$

As before with Thevenin, if we short-out the two voltage sources and open circuit terminals A and B, the two resistors are now effectively connected together in parallel. The value of the equivalent resistor R_{EQ} is found by calculating the parallel resistance combination across terminals A and B as shown.

$$R_{EQ} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{10 \times 20}{10 + 20} = \frac{200}{30} = 6.67 \Omega$$

Thus Norton's equivalent circuit consists of an equivalent current source, I_{EQ} of 2A connected in parallel with an equivalent resistance, R_{EQ} of 6.67Ω . We can now connect the 40Ω load resistor back into Norton's equivalent circuit as shown in Figure 18.

FIGURE 18. NORTON EQUIVALENT CIRCUIT



We can see that the two resistors, R_{EQ} and R_L are now connected together in parallel across the circuit terminals of A and B. We can now calculate the equivalent parallel resistance, (R_T) as well as the voltage across and the current flowing through the external 40Ω load resistor.

$$R_T = \frac{R_{EQ} \times R_L}{R_{EQ} + R_L} = \frac{6.67 \times 40}{6.67 + 40} = \frac{266.8}{46.67} = 5.72 \Omega$$

The voltage developed across terminals A – B:

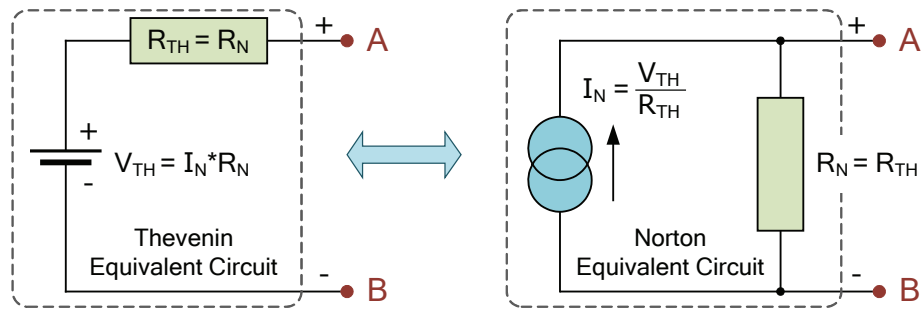
$$V_{A-B} = I_{EQ} \times R_T = 2 \times 5.72 = \underline{11.44 \text{ volts}} \quad (\text{the same voltage value})$$

The current flowing in R_L :

$$I_L = \frac{V_{A-B}}{R_L} = \frac{11.44}{40} = \underline{0.286 \text{ Amperes}}$$

(the same current value as for the previous analysis)

Conversion between Thevenin and Norton Equivalent Circuits



Where: V_{TH} = Thevenin Voltage, R_{TH} = Thevenin Resistance, V_N = Norton Voltage, I_N = Norton Current, and R_N = Norton resistance.

11. MAXIMUM POWER TRANSFER THEOREM

Maximum Power Transfer Theorem is another useful circuit analysis method to ensure that the maximum amount of power will be dissipated in the load resistance when the value of the load resistance is exactly equal to the internal resistance of the source.

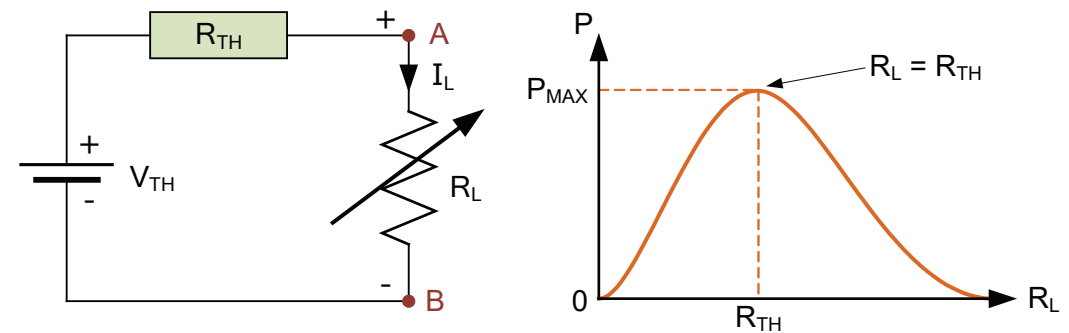
That is: **“the maximum amount of power will be dissipated in the load resistance if it is equal in value to the Thevenin or Norton source resistance of the linear network supplying the power”.**

However, if the circuit is a voltage source such as a Thevenin equivalent circuit, then for maximum power transfer, the load resistance must be equal to the Thevenin series resistance, R_{TH} .

If the supplying circuit is a current source such as a Norton equivalent circuit, then the load resistance must be equal to the internal parallel resistance, R_N for maximum power transfer to occur.

Consider the previous Thevenin equivalent circuit with an externally connected variable resistor across its output terminals A-B as shown in Figure 19.

FIGURE 19. THEVENIN'S CIRCUIT



Circuit current:
$$I_L = \frac{V_{TH}}{R_L + R_{TH}}$$

The power consumed by the load is:
$$P_L = I_L^2 \times R_L = \left(\frac{V_{TH}}{R_L + R_{TH}} \right)^2 \times R_L$$

For maximum power transfer to occur between source and load, the source and load resistances must be of equal value. That is $R_L = R_{TH}$

$$\frac{dP_L}{dR_L} = (R_L - R_{TH}) = 0$$

Therefore maximum power to the load is produced when:

$$P_{L(MAX)} = \frac{V_{TH}^2 \times R_L}{4R_L^2} = \frac{V_{TH}^2}{4R_L}$$

So clearly, maximum power is transferred from the source to the load only when the load impedance is equal in value to the source impedance (resistive or complex). The process of making the two impedances equal is called impedance matching.

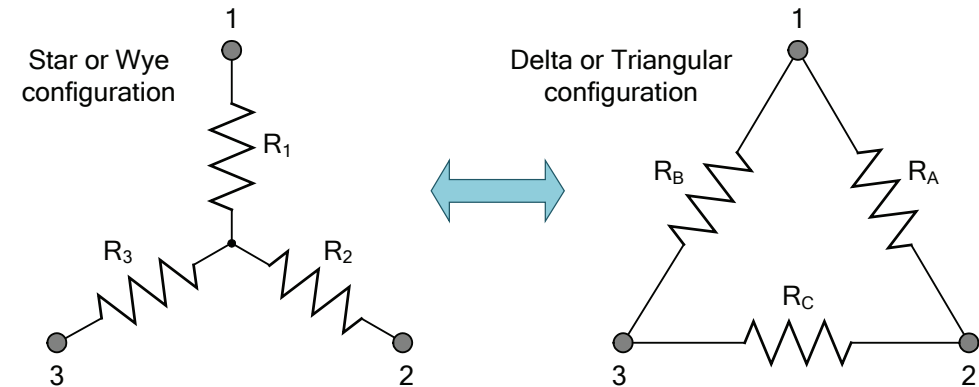
13. STAR/DELTA AND DELTA/STAR TRANSFORMATIONS

Resistive branches of an electrical network can be connected in a number of different ways and there are instances in which the resistances are neither connected together in series nor in parallel. The most common example is a three terminal network where the resistances are connected in either Star (Wye) or Delta (Triangular) form.

In a Delta or Triangular connection, the three branches are so connected, that they form a closed loop where the end of one resistor connects to the start of next resistor. When either end of the three resistors are connected together to a common junction or point to form a Y-shape, then this configuration is known as Star or Wye connection.

Just as it is possible to simplify series and parallel circuits, it is also possible to convert delta networks into star and vice-versa simply by using series parallel circuit techniques. Thus a Star-Delta Transformation can be converted to the Delta-Star Transformation as shown.

FIGURE 20 STAR AND DELTA CIRCUITS



Delta - Star Transformation

$$R_1 = \frac{R_A R_B}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_B R_C}{R_A + R_B + R_C}$$

Star - Delta Transformation

$$R_A = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3}$$

$$R_B = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

$$R_C = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

[End of this DC Circuit Theorems eBook](#)

Last revision: March 2022

Copyright © 2022 Aspencore

<https://www.electronics-tutorials.ws>

Free for non-commercial educational use and not for resale

With the completion of this DC Circuit Theorems eBook you should have gained a basic understanding and knowledge of electrical circuit analysis. The information provided here should give you a firm foundation for continuing your study of electronics and electrical engineering. In ebook 4 we will learn about the different types of Resistive Circuits.

For more information about any of the topics covered here please visit our website at:

[*www.electronics-tutorials.ws*](https://www.electronics-tutorials.ws)

Main Headquarters

245 Main Street
Cambridge, MA 02142

www.aspencore.com

Central Europe/EMEA

Frankfurter Strasse 211
63263 Neu-Isenburg, Germany

info-europe@aspencore.com